

## LAGRANGE CALLAWAY, GA, USA

WMO: 747807

Lat: 33.017N

Lon: 85.067W

Elev: 208

StdP: 98.85

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 03821

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.9       | -3.3 | -13.0                         | 1.2 | -1.0 | -11.1 | 1.5 | 1.9  | 9.0                   | 7.7  | 8.2 | 7.1  | 1.2                   | 340  | 0.359 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.3                   | 35.1            | 25.0 | 33.9 | 24.5 | 32.8 | 24.4 | 27.5                | 30.7 | 26.8 | 30.3 | 26.2 | 29.9 | 2.7                  | 260  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.8                            | 22.9 | 28.6 | 26.1 | 22.0 | 28.4 | 25.2 | 20.9 | 27.9 | 88.2          | 31.3 | 85.2 | 30.6 | 82.5 | 29.5 | 32.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.5           | 6.1  | 5.2 | DB                         | -8.8 | 36.7               | 2.5 | 1.9                                                | -10.6 | 38.1       | -12.0 | 39.2       | -13.4 | 40.3       | -15.2 | 41.7 | (4) |
| (5)               |      |     | WB                         | -9.2 | 27.7               | 2.7 | 1.0                                                | -11.2 | 28.4       | -12.8 | 29.0       | -14.3 | 29.5       | -16.3 | 30.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.3   | 6.9   | 9.1   | 13.2  | 16.7  | 21.1  | 25.3  | 26.5  | 26.5  | 23.6  | 17.7  | 11.6  | 8.6   | (6)  |  |
|                                                                        | DBStd       | 8.03   | 5.42  | 5.05  | 4.65  | 3.76  | 3.45  | 2.12  | 1.77  | 1.96  | 2.96  | 4.48  | 4.43  | 4.96  | (7)  |  |
|                                                                        | HDD10.0     | 347    | 127   | 71    | 23    | 3     | 0     | 0     | 0     | 0     | 0     | 3     | 31    | 89    | (8)  |  |
|                                                                        | HDD18.3     | 1450   | 354   | 260   | 168   | 73    | 17    | 0     | 0     | 0     | 2     | 67    | 207   | 302   | (9)  |  |
|                                                                        | CDD10.0     | 3003   | 32    | 47    | 121   | 204   | 344   | 459   | 512   | 511   | 407   | 241   | 79    | 47    | (10) |  |
|                                                                        | CDD18.3     | 1064   | 1     | 3     | 8     | 24    | 102   | 209   | 254   | 252   | 159   | 46    | 5     | 2     | (11) |  |
|                                                                        | CDH23.3     | 10153  | 2     | 13    | 99    | 328   | 1027  | 2040  | 2468  | 2377  | 1364  | 396   | 36    | 4     | (12) |  |
|                                                                        | CDH26.7     | 4363   | 0     | 1     | 12    | 59    | 362   | 948   | 1177  | 1134  | 562   | 105   | 4     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 1.7    | 2.1   | 2.2   | 2.2   | 2.0   | 1.7   | 1.4   | 1.3   | 1.2   | 1.3   | 1.5   | 1.6   | 1.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1344   | 130   | 127   | 155   | 104   | 99    | 107   | 128   | 96    | 94    | 83    | 116   | 112   | (15) |  |
|                                                                        | PrecMax     | 1693   | 247   | 217   | 277   | 186   | 311   | 223   | 351   | 176   | 185   | 254   | 305   | 253   | (16) |  |
|                                                                        | PrecMin     | 808    | 62    | 43    | 34    | 47    | 17    | 35    | 44    | 40    | 20    | 12    | 39    | 34    | (17) |  |
|                                                                        | PrecStd     | 218    | 46    | 45    | 63    | 36    | 63    | 58    | 72    | 36    | 47    | 65    | 61    | 46    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 23.0  | 25.6  | 27.9  | 30.0  | 33.2  | 36.1  | 36.3  | 37.1  | 35.0  | 31.8  | 27.0  | 24.0  | (19) |  |
|                                                                        |             | MCWB   | 18.1  | 19.3  | 18.2  | 21.2  | 22.9  | 24.8  | 25.8  | 25.0  | 23.3  | 23.1  | 19.3  | 19.8  | (20) |  |
|                                                                        | 2%          | DB     | 20.4  | 22.3  | 26.1  | 28.1  | 31.9  | 34.1  | 34.8  | 35.0  | 33.4  | 29.2  | 23.9  | 21.8  | (21) |  |
|                                                                        |             | MCWB   | 16.8  | 17.5  | 18.1  | 19.5  | 22.6  | 24.3  | 25.5  | 25.0  | 23.6  | 21.6  | 18.8  | 18.9  | (22) |  |
|                                                                        | 5%          | DB     | 18.1  | 20.2  | 23.9  | 27.2  | 30.3  | 32.9  | 33.5  | 33.7  | 32.0  | 27.5  | 22.3  | 19.5  | (23) |  |
|                                                                        |             | MCWB   | 15.7  | 16.6  | 17.0  | 18.9  | 22.1  | 24.2  | 25.1  | 24.7  | 23.5  | 20.4  | 17.6  | 17.3  | (24) |  |
|                                                                        | 10%         | DB     | 16.3  | 18.2  | 22.2  | 25.4  | 28.8  | 32.1  | 32.4  | 32.5  | 30.2  | 25.7  | 20.3  | 17.6  | (25) |  |
|                                                                        |             | MCWB   | 13.5  | 14.4  | 16.1  | 17.9  | 21.7  | 24.1  | 25.0  | 24.8  | 23.4  | 19.7  | 16.4  | 15.5  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 19.8  | 20.8  | 21.5  | 24.1  | 25.7  | 27.9  | 28.5  | 28.0  | 26.9  | 25.1  | 23.2  | 21.9  | (27) |  |
|                                                                        |             | MCDB   | 21.4  | 22.6  | 24.0  | 27.6  | 29.0  | 31.9  | 33.4  | 31.0  | 30.4  | 27.1  | 23.8  | 22.5  | (28) |  |
|                                                                        | 2%          | WB     | 18.2  | 19.4  | 20.2  | 22.1  | 24.5  | 26.8  | 27.5  | 27.3  | 25.9  | 23.7  | 21.2  | 19.9  | (29) |  |
|                                                                        |             | MCDB   | 19.6  | 21.3  | 23.3  | 24.9  | 28.3  | 30.6  | 31.0  | 30.3  | 29.6  | 26.7  | 22.4  | 21.2  | (30) |  |
|                                                                        | 5%          | WB     | 17.0  | 17.9  | 18.9  | 20.8  | 23.7  | 26.0  | 26.8  | 26.6  | 25.1  | 22.6  | 19.3  | 18.1  | (31) |  |
|                                                                        |             | MCDB   | 18.1  | 19.6  | 22.4  | 24.5  | 27.7  | 30.1  | 30.3  | 30.1  | 28.4  | 25.7  | 21.4  | 18.9  | (32) |  |
|                                                                        | 10%         | WB     | 14.3  | 16.0  | 17.5  | 19.6  | 22.9  | 25.3  | 26.1  | 26.0  | 24.2  | 21.6  | 17.5  | 16.1  | (33) |  |
|                                                                        |             | MCDB   | 15.4  | 17.6  | 20.7  | 23.6  | 27.0  | 29.7  | 29.8  | 29.9  | 27.7  | 24.4  | 19.3  | 17.6  | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 12.1  | 12.5  | 13.6  | 14.4  | 13.2  | 12.1  | 11.3  | 11.2  | 11.7  | 13.1  | 13.9  | 11.8  | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 13.7  | 13.9  | 15.7  | 16.2  | 14.9  | 14.2  | 13.0  | 13.4  | 13.8  | 14.9  | 15.2  | 12.7  | (36) |  |
|                                                                        |             | MCWBR  | 9.8   | 9.2   | 8.3   | 8.0   | 6.4   | 5.4   | 4.9   | 4.8   | 5.0   | 7.1   | 9.3   | 9.7   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 11.2  | 10.8  | 12.8  | 12.2  | 11.7  | 11.7  | 11.4  | 11.2  | 10.8  | 10.6  | 11.1  | 10.8  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.319 | 0.322 | 0.354 | 0.387 | 0.447 | 0.480 | 0.522 | 0.524 | 0.423 | 0.371 | 0.344 | 0.323 | (40) |  |
|                                                                        | taud        |        | 2.524 | 2.509 | 2.443 | 2.333 | 2.207 | 2.169 | 2.062 | 2.045 | 2.319 | 2.450 | 2.465 | 2.527 | (41) |  |
|                                                                        | Ebn at Noon |        | 901   | 934   | 924   | 902   | 847   | 815   | 779   | 770   | 841   | 867   | 860   | 875   | (42) |  |
|                                                                        | Edh at Noon |        | 84    | 94    | 109   | 126   | 145   | 150   | 166   | 167   | 122   | 99    | 89    | 80    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.76  | 3.31  | 4.42  | 5.58  | 6.15  | 6.13  | 5.97  | 5.46  | 4.78  | 4.07  | 3.08  | 2.35  | (44) |  |
|                                                                        | RadStd      |        | 0.16  | 0.40  | 0.38  | 0.38  | 0.50  | 0.49  | 0.40  | 0.32  | 0.45  | 0.52  | 0.30  | 0.25  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (47 neighbors) |  | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +156    | +106    |

Nomenclature: See separate page